Supplementary Material: V-SyncLock: Signed Vertical Motion for Sub-Frame Synchronization and Drift Correction of Consumer Camera Rigs

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This supplement records how every design choice was made, gives the full tables behind the paper's summaries, and lists how each number is reproduced. "Paper" references (e.g., paper Sec. 6) point to the main text; the research questions RQ1–RQ4 are those of the paper. Unless stated otherwise, "V-SyncLock input" is per-joint speed plus signed velocity along the body axis, "image vertical" the same with the image's vertical axis, and "speed" the per-joint speed baseline; errors are in frames at 30 fps.

S1 Selection Protocol and Decisions

Each tunable choice was decided on development data by a rule written down before the development result was computed, and the test splits were then evaluated once with the selected value (Table S1). We call this development-driven selection: for some choices, earlier exploration of test data preceded the rule, as listed under "Order of evaluation" below. The full log, including rules whose outcome was negative, is part of the released code.

Order of evaluation. For four choices a test result existed before the development rule was written. Clip length and placement method had been examined on the TotalCapture test split with the speed baseline; the development rules selected the same 10-s context and gated least squares. The body-axis rule was first checked on test samples (TotalCapture / AthletePose3D / SportsPose +0.04 / +0.00 / −0.04) and then confirmed on development data (row above; image-axis results in Sec. S3). Its AthletePose3D development subjects (S3–S4) are also among the 233 reported AthletePose3D trials; on S2 alone (74 trials, used for no decision) speed CC 2.78 becomes 0.17 with the final input (CSF 0.19, DTW 0.13, Caspi-style 0.17), as reported in the caption of paper Table 5. For the periodicity router the development rule selected τ=0.40 although the test analysis had used τ=0.5; 0.40 is the value reported. The two-stage router's accuracy point was fixed on development data after the cost point had been tested (Sec. S6). A SportsPose run with the image axis first used sensor rather than display-upright axes (the cameras are mounted rotated by ±90°; 3.52 frames) and was corrected to the definition fixed in advance.

S2 Motion Signals

The Pose2Sim-style signal follows `synchronization.py` [1] (Butterworth 4th order, 6 Hz, sum of absolute vertical speed over all joints); the OpenCap-style signal follows the general mode of `utilsSync.py` [2] (mean signed vertical velocity of the two ankles). Both are re-implemented as signals under our correlator, not run end-to-end: Pose2Sim's window around maximum speed and OpenCap's reprojection refinement and activity-specific modes are omitted. By activity on TotalCapture (sub readout), the OpenCap-style signal gives walking 2.10, acting 2.17, freestyle 2.58; speed + $\dot{y}$ gives 0.68, 0.26, 0.39. On SportsPose, speed + $\dot{y}$ gives jump 0.13, soccer 0.62, tennis 1.14, baseball 0.67, volleyball 0.69.

S3 Image-Vertical versus Body-Axis Input

With the image-vertical axis instead of the body axis (cameras upright), the main results change little: TotalCapture 10-s test CC 0.441 / CSF 0.385 (body axis 0.483 / 0.432); AthletePose3D 0.182 / 0.344 (0.184 / 0.362); SportsPose 0.648 / 0.612 (0.605 / 0.578); gated least-squares placement 0.31 on TotalCapture and 0.50 on SportsPose (0.32 / 0.52); TotalCapture 3D at three frames of drift CC-c2f 9.4 / CSF-c2f 8.5 mm (9.7 / 9.0); Panoptic 3D 0.90 / 0.89 mm (0.89 / 0.65); our capture, three frames of drift, 0.43 / 0.33 (0.47 / 0.39). The image axis is slightly better on TotalCapture and AthletePose3D, the body axis on SportsPose and in Panoptic 3D; only the body axis is invariant to roll.

S4 Panoptic Controlled Benchmark

Under simulated noise the input brings no gain in drift error, while 3D error still falls slightly. On constant offsets the body-axis input helps CSF and hurts CC; the clips (people standing and talking) contain little vertical motion, and the image-vertical input gives CC 0.180.

S5 TotalCapture Details

Clip length. Test, pairs ≤60°, mean error (share >1) for CC / CSF: walking 0.62 / 0.58 (2 s), 0.77 / 0.65 (5 s), 0.83 / 0.70 (10 s); acting 0.59 / 0.63, 0.29 / 0.33, 0.22 / 0.24; freestyle 0.72 / 0.75, 0.58 / 0.51, 0.53 / 0.39. Longer context helps acting and freestyle; for walking CC degrades with length (more equally good alignments one step apart) while CSF does not.

Viewpoint. Share of estimates >1 frame by subject-centred angle (0–30, 30–60, 60–75, 75–90, 90–105, 105–120, 120–150°): CC 4, 11, 9, 10, 14, 5, 7%; CSF 2, 9, 10, 7, 8, 7, 7%. SportsPose (same bins plus 150–180°): CC 8, 14, 19, 28, 20, 17, 16, 8%; CSF 9, 10, 22, 27, 14, 16, 17, 10%.

Gate. Self-calibrated null at q = 0.90 / 0.95 / 0.99 keeps 64 / 60 / 52% of the 10-s pairwise estimates with precision 96 / 96 / 97% and 68 / 73 / 81% of the failures removed (CC; CSF 65 / 60 / 51% kept). Walking: 85% of estimates correct, 23 / 17 / 11% kept. A phase-fixed null (lag-0 correlation of windows ≥3 s apart, threshold at $q^{1/6}$), meant to stop the maximum over lags from inflating the walking null, failed its rule on development (walking kept must rise by ≥10 points): walking kept fell from 18 to 13% (threshold 0.352→0.377), so steady gait correlates across windows even at a fixed lag.

Placement. On 10-s development clips the mean placement error is 0.73 frames for the star, 1.00 for the spanning tree, 0.64 for least squares over all edges, 0.72 for the gated tree and 0.55 for gated

Table S1: Decisions, the rule fixed in advance, the development outcome and what was adopted. TC: TotalCapture; dev: S1–S3 walking/acting/freestyle 1.

Choice	Rule (fixed in advance)	Development outcome	Adopted / test
Motion input	lowest TC dev error among six formulations (speed; $\dot{y}$; speed + $\dot{y}$; speed + L–R speed; speed + L–R $\dot{y}$; speed + $\dot{y}$ + L–R $\dot{y}$)	1.44 / 0.64 / **0.59** / 1.11 / 1.28 / 0.64	speed + signed $\dot{y}$; test Table S2
Vertical axis	body axis replaces image axis if within 0.05 frames at zero roll	dev: TC CC −0.019 [−0.084, +0.052], CSF −0.003; AthletePose3D S3–S4 CC +0.003, CSF +0.026	body axis; Table S3
Clip length (TC)	lowest dev error per estimator; ties (<0.02) to the shorter	CC 0.79 / 0.62 / **0.51**, CSF 0.89 / 0.54 / **0.46** (2/5/10 s)	10 s for both
Placement	least squares if its dev error over placed cameras is below the gated tree's	0.55 vs. 0.72 (10 s)	least squares
Router	cheapest δ with dev error within 0.05 of CSF alone; grid 0.25–3.0	every δ qualifies; δ=3 (grid edge); argmin δ=0.75 (0.433), fixed after the δ=3 test run	two-stage; test 0.432 at 4% / 0.412 at 25% CSF
Edge pruning	keep k strongest edges per camera if dev error rises ≤0.05	+0.09 / +0.10 (k = 3 / 2; speed input)	not adopted
Coarse-to-fine prior	Panoptic validation sequence only	supported prior beats long-window prior; ρ = 2, 3 tie	supported, ρ=3
Clock-rate prior	fixed scale 2 frames/min, no tuning	–	streaming only (Sec. S7)

Table S2: Motion signals under the same cross-correlation, integer-lag (int) and parabolic (sub) readout; mean error per sequence/trial. Paired differences of speed + signed $\dot{y}$ (sub) to every row exclude zero except $\dot{y}$ (sub) on TC (−0.05 [−0.10, 0.00]).

	TC 10 s		AP3D		SP	
Signal	int	sub	int	sub	int	sub
$\sum \lvert\dot{y}\rvert$ (Pose2Sim-style)	3.17	3.15	2.95	2.89	1.96	1.92
ankle mean $\dot{y}$ (OpenCap-style)	2.31	2.29	1.52	1.50	3.05	3.03
per-joint $\lvert\dot{y}\rvert$	2.60	2.57	3.41	3.34	1.64	1.59
per-joint speed	1.35	1.32	3.27	3.24	1.06	1.00
per-joint $\dot{y}$	0.55	0.49	0.27	0.13	1.08	1.03
speed + $\dot{y}$	0.50	0.44	0.30	0.18	0.71	0.65

Table S3: Camera roll of one view: mean error (frames).

	Input	0	5	10	15	20	30	45	60	90
TC	image $\dot{y}$	0.44	0.45	0.46	0.48	0.50	0.56	0.70	0.92	1.77
	body u	0.48	0.48	0.48	0.48	0.49	0.49	0.49	0.49	0.48
AP3D	image $\dot{y}$	0.18	0.18	0.19	0.19	0.20	0.22	0.29	0.48	3.13
	body u	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.18
SP	image $\dot{y}$	0.65	0.63	0.62	0.62	0.63	0.67	0.81	1.06	1.76
	body u	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61

Table S4: Panoptic constant offsets (1,000 pairs each), mean error in frames (share within 0.1 frame). Speed: speed baseline (CSF trained on speed).

	Clean, test		Noise, validation	
Method	speed	V-SyncLock input	speed	V-SyncLock input
CSF	0.106 (61%)	**0.080** (73%)	0.411	**0.345**
CC	0.148 (63%)	0.259 (75%)	0.620	0.521
DTW	0.153 (57%)	0.153 (57%)	0.686	0.686

least squares (72% of cameras placed). On 2-s test clips the same five methods give 0.73, 0.77, 0.51, 0.59 and 0.50 (76% placed).

Clock-rate prior offline. With the speed input, most of the coarse-to-fine error without drift was slope noise, and the prior lowered

Table S5: Panoptic drift recovery, E_{drift} (frames), V-SyncLock input; speed input in parentheses at three frames.

	Clean, 8 clips, d				Noise, 12 clips
Method	0	1	2	3	d=3
CC-const	0.037	0.264	0.533	0.799 (0.841)	0.866 (0.987)
Caspi-style	0.303	0.285	0.242	0.273 (0.273)	0.362 (0.362)
CC-slide	0.072	0.058	0.056	0.085 (0.421)	1.246 (1.253)
CC-c2f	0.063	0.055	0.050	0.063 (0.134)	0.398 (0.381)
CSF-slide	0.045	0.048	0.039	0.050 (0.127)	0.367 (0.386)
CSF-c2f	0.045	0.048	0.039	0.050 (0.121)	0.373 (0.263)

Table S6: Panoptic 3D, MPJPE (mm) after correction, V-SyncLock input (speed input in parentheses). Clean: 11 clips incl. the validation sequence; test only: 4 clips; noise: 14 clips.

	Clean		Test only	Noise
Method	d=0	d=3	d=3	d=3
no correction	3.65	13.20	7.71	14.13
CC-const	1.21	5.00 (4.93)	2.20 (2.19)	10.09 (10.16)
CC-c2f	1.02	0.89 (2.88)	0.21 (0.31)	9.14 (10.37)
CSF-c2f	0.80	0.65 (1.51)	0.27 (0.23)	9.13 (9.30)
true timing	0.37	0.36	0.24	8.62

Table S7: TotalCapture drift and 3D (48 10-s test clips), MPJPE (mm) to the genlocked triangulation. "+prior": slope shrunk by the clock-rate prior (2 frames/min).

d	none	CC-const	Caspi	CC-c2f	CSF-c2f	CC+pr.	CSF+pr.	true
V-SyncLock input								
0	11.9	7.6	28.1	8.4	9.0	7.3	7.3	3.3
1	23.0	9.2	28.4	9.0	8.2	9.5	7.9	1.9
2	33.8	13.5	29.4	9.8	8.8	13.1	11.7	1.8
3	44.3	18.0	29.9	9.7	9.0	17.0	15.8	2.0
speed input								
0	11.9	18.3	28.1	16.5	17.3	9.5	11.8	3.3
3	44.3	29.3	29.9	18.3	17.6	20.8	19.8	2.0

3D error from 16.5 to 9.5 mm (−2.4 mm [−4.0, −0.5] versus no correction). With the V-SyncLock input the slope is already well determined and the prior only costs accuracy at larger drift (Table S7);

Table S8: Routing on TotalCapture 10-s test clips, V-SyncLock input. Cost relative to CC on whole 10-s clips (CC 1.19, quarters 3.60, CSF 45.4 ms).

Rule	Error	>1	CSF calls	Cost
CC only	0.483	9%	0%	1.0
CSF only	0.432	8%	100%	38
periodicity, τ=0.40	0.445	8%	77%	29
two-stage, $D \geq 3$	0.432	8%	4%	5.5
two-stage, $D \geq 0.75$	0.412	7%	25%	6.5

V-SyncLock therefore uses it only when streaming. A hard slope test (slope kept only if $|\hat{s}|/\mathrm{se} \geq 1.96$, otherwise a constant offset), meant to remove the small tilt of CSF lines on our nearly drift-free capture, failed its development rule: it helped without drift (E_{drift} CC 0.359→0.335, CSF 0.373→0.307) but cost up to 0.07 frames at d = 1–3, as did a bootstrap test of the slope with the speed baseline, because on 10-s clips the slope's standard error does not separate moderate drift from noise.

S6 Routing, Pruning and Cost

Three routing rules were tested on 10-s TotalCapture clips, each fixed before its development result. With the speed baseline, routing on an ambiguous CC peak (margin between the highest and the second peak) came no closer than 1.19 vs. 1.10 frames to CSF on development (23% CSF calls) and was not run on test; the temporal consistency D of the paper reached 1.034 on test (CSF 0.974, CC 1.320) at 19% CSF calls. On the development split (speed input), D also predicts the clips on which CSF beats CC by more than half a frame best: AUC 0.78, against 0.74 for CC on the two halves of the clip, 0.69 for the peak margin and 0.58 for the input's periodicity; the margin misses clips where CC is confidently wrong, with an aliased peak that is sharp but moves between quarters. With the V-SyncLock input, D is the paper's estimator: test 0.432 at 4% CSF calls (Table S8). Its accuracy point, the development argmin δ=0.75, was written down after this test run and evaluated once: 0.412 at 25% CSF calls, −0.033 [−0.058, −0.014] vs. the periodicity router; by activity (two-stage / CC / CSF) walking 0.558 / 0.665 / 0.592, acting 0.261 / 0.259 / 0.276, freestyle 0.419 / 0.547 / 0.424. Keeping only the k strongest edges per camera after pass 1 (union over cameras) changed placement error on test by +0.03 / 0.00 frames (k = 2 / 3; 80% / 81% of cameras placed, all edges 0.74 and 81%) but on development by +0.10 / +0.09 (speed input), so it was not adopted. Its cost effect, measured with the V-SyncLock input on one minute of eight-camera video (one thread): k=2 keeps 9 of 28 pairs and cuts pass 2 from 13.6 to 5.0 s with CC and from 110 to 34 s with CSF.

Cost breakdown. Shared stages for eight cameras, seconds per minute of video on one thread (paper Table 2, 18.8 s in total): coarse alignment 0.06, pass 1 4.3, self-calibrated null 14.1, IRLS placement <0.01, resampling 0.01, triangulation 0.29. MediaPipe Pose takes 59 s per camera.

The threshold is a cost–accuracy trade-off on TotalCapture rather than a tuned optimum (Fig. S1): every δ on the grid is within 0.05 frames of CSF alone, so the rule selects the cheapest. On our optically timed capture the same δ does not transfer fully (router 0.90 vs. CSF 0.88 frames). Applied unchanged to the short clips of the other datasets it stays at CC's level: AthletePose3D 0.188 at δ=3 (1% CSF calls) and 0.193 at δ=0.75 (2%), CC 0.184, CSF 0.362; SportsPose 0.593 (6%) and 0.568 (49%; −0.037 [−0.074, −0.003] vs. CC), CC 0.605, CSF 0.578. On sequence-held-out TotalCapture data both points keep their properties (Sec. S10).

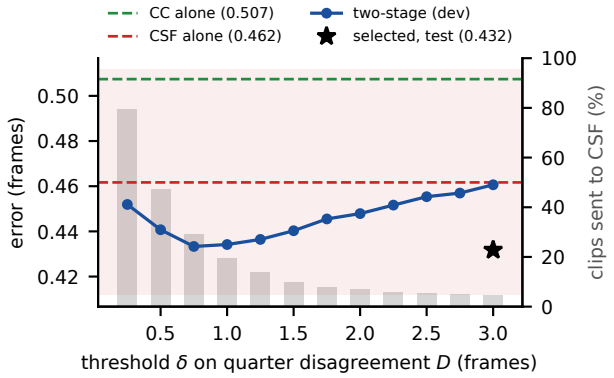


Figure S1: Two-stage router on the TotalCapture development split: error (line) and share of clips sent to CSF (bars) for every threshold of the pre-registered grid; shaded: within 0.05 frames of CSF alone. The rule picks the cheapest qualifying δ, here the grid edge; the star is its single test result. Intermediate thresholds (δ = 0.75–1.25, 14–29% CSF calls) beat both estimators on development data (0.433–0.436); the minimum, δ=0.75, is the accuracy point.

S7 Streaming

All eight TotalCapture cameras are replayed in one-second chunks (15 test sequences, first 120 s; in practice 50–80 s); each non-reference camera gets an offset uniform in ±1.5 frames and a drift uniform in ±3 frames/min, applied exactly by cubic resampling of the 60-fps detections. Table S9 gives the mean offset error of placed cameras by elapsed time. Nothing is corrected before the first fit at 10 s. An earlier four-camera replay with drift of 1 or 3 frames/min shows the two regimes of Eq. (**??**). An unregularized causal line fitted on 10–20 s swings between −4 and +2 frames/min (18.7 mm at 1 frame/min). A constant offset is better early but falls 2.21 frames behind after a minute at 3 frames/min (30.1 mm). The shrunk line beats both (13.4 and 20.7 mm). A refit of all 28 pairs takes 2.2 s on average, 3.1 s at the 95th percentile and 3.9 s at most, per 5 s of video on one CPU thread. These numbers include the gate-before-update rule and the ±6 frames/min slope bound (main paper, Sec. 4); without them causal errors were 0.68 / 0.51 / 0.35 and 0.28 / 0.33 / 0.30 with the prior. Four remedies, in three variants, for the failure seen on our capture (misleading early motion that passes the gate and biases later, path-dependent fits) were tested on the development split with a rule written before each result: the streaming stage of the paper must not be worse by more than 0.02 frames on the unmodified replay and at drifts of 0, ±1 and ±3 frames/min, and must improve a misleading 30-s lead-in (each camera showing real motion from a different time) or a rig with two cameras stopping at

Table S9: Streaming on the replayed eight-camera TotalCapture rig: mean offset error (frames) of placed cameras by elapsed time, and share of cameras placed. Nothing is corrected before the first fit at 10 s.

Mode	0–10 s	10–30 s	30–60 s	60 s+	placed
no correction	0.80	0.97	1.40	2.54	–
streaming	–	0.39	0.37	0.39	69%
streaming + clock prior	–	0.30	0.33	0.35	69%
offline (whole sequence)	0.31	0.23	0.30	0.27	86%

40 s by at least 0.05 frames. (i) Re-centring cached windows whose lag centre moved by more than ρ frames, together with excluding cameras that stopped recording from the placement of the others (one variant): lead-in 1.075 → 1.085 / 1.000 → 1.135 (no prior / prior), with over 2,000 recomputed windows per sequence; stopped cameras 0.397 → 0.447 / 0.343 → 0.367. (ii) Hough voting of window profiles over (offset, slope) with acquisition and tracking states: lead-in 1.375 / 1.255, and worse on the unmodified replay (0.383 → 0.437); offline it was marginally better (0.23–0.25 vs. 0.24–0.27). (iii) Pass-1 windows kept at the coarse alignment with a ±30-lag search, 50 null samples before a line is accepted, and a motion floor: worse in every scenario (unmodified replay 0.38 → 0.42, lead-in 1.08 → 2.58, no prior), because the wide search on 80 frames admits more periodic false peaks and misleading motion still passes a gate with 50 null samples. None was adopted. 3D error is not compared across modes because they place different sets of cameras.

Clock-rate prior when streaming. An earlier causal experiment, run before the gate-before-update rule and the slope bound existed, shows the two regimes of the prior (paper Eq. 14). Cameras 8, 4 and 7 of the 15 test sequences drift by r = 1 or 3 frames/min against camera 1; each pair's line to the reference is refitted every 5 s on all past windows. Mean offset error at 0–10 / 10–30 / 30–60 / 60+ s and MPJPE (mm, frames after 10 s) to the genlocked triangulation, for r=1: uncorrected 0.58 / 0.83 / 1.20 / 1.58, 25.3 mm; causal line without prior 0.58 / 0.80 / 0.62 / 0.44, 18.7 mm (slopes of −4 to +2 frames/min from 10–20 s of history); causal constant offset 0.58 / 0.51 / 0.71 / 1.00, 15.3 mm; causal line with prior – / 0.47 / 0.58 / 0.42, 13.4 mm; offline 10.9 mm. For r=3: uncorrected 48.8 mm; without prior 22.0 mm; constant offset 0.75 / 0.93 / 1.57 / 2.21, 30.1 mm; with prior 20.7 mm; offline 14.6 mm. The constant offset (an exploratory variant added after the first run) is best early and worst late; the prior is best overall.

S8 Our Captures

First capture. After timestamp resampling and coarse alignment (146-s performance segment, reference Phone A), reprojection error over all / fast frames: coarse only 8.26 / 10.02 px; CC-const 8.10 / 9.52; Caspi-style 8.13 / 9.62; CC-slide 8.10 / 9.51; CC-c2f 8.10 / 9.54; CSF-slide 8.10 / 9.51; CSF-c2f 8.08 / 9.48. Shifting all non-reference cameras by 0.5, 1, 2 and 3 frames raises the fast-frame error from 10.0 to 10.6, 12.1, 15.5 and 19.0 px, so the criterion detects gross errors but cannot rank sub-frame methods.

Second capture read by frame index. Read as frames at the nominal 30 fps instead of on timestamps, the devices of the second capture drift relative to the reference by −0.91 (Phone B, variable frame rate, index 7 frames behind its clock over the take), −0.04 (Phone C), −0.12 (Webcam 0) and −60.5 frames/min (Webcam 1, 28.94 fps). E_{drift} for single lag / CC-c2f / CSF-c2f, with the error of the best constant and the best line fitted to the truth: Phone B 3.77 / 1.92 / 2.00 (3.06, 1.78); Phone C 0.15 / 0.10 / 0.18 (0.09, 0.01); Webcam 0 0.74 / 0.38 / 0.27 (0.18, 0.19); Webcam 1 about 110 for all (2.09), beyond the search.

Table S10: Which configuration in which regime (signed input; errors in frames unless noted). TC: TotalCapture; AP3D: AthletePose3D; SP: SportsPose.

Regime	Evidence	Best	Note
short clip, sprint	AP3D	DTW (0.13), CC (0.18)	CSF 0.36
3-s sports trial	SP	CC ≈ CSF (0.61 / 0.58)	75–90° pairs hardest
10-s everyday motion	TC	two-stage (0.43)	CSF on 4% of clips
constant offset, standing	Panoptic	CSF (0.08)	CC worse than with speed
drift ≤ 1 frame, real det.	ours	single lag (0.19–0.28)	c2f 0.27–0.42
drift 3 frames, real det.	TC, ours	CSF-c2f (0.40, 0.39)	3D 9.0 mm vs. 44 uncorr.
detector noise	Panoptic	CC-c2f ≈ CSF-c2f (0.40 / 0.37)	no gain from signed input
rig, views that disagree	TC, SP	gated LS (0.32, 0.52)	19–39% abstain; no gain on S10
rig, all views observable	AP3D	any placement (0.16)	gate keeps all
streaming, ≤30 s history	TC rig	+ clock-rate prior (0.30)	0.39 without prior
real clocks, everyday actions	ours (torch)	CSF (0.88); rig CC (0.18)	streaming 0.48–1.0
crawling, lying	ours (torch)	CSF (0.50)	CC 1.06; 0.67 post hoc

Third capture (clap-aligned). Five devices (one iPhone at 3840×2160, three at 1920×1080, one at 1280×720; 29.60–30.02 fps) had been trimmed to a common frame by eye on a hand clap; the trims (553, 33, 222, 102, 229 frames) were recovered exactly by image matching. On timestamps, offsets from CC-c2f differ from the clap alignment by per-camera constants of 0, −4.57, +0.31, −1.84 and −1.13 frames (CSF-c2f: 0, −4.56, +0.29, −1.91, −1.18), with residual RMS 0.07 frames, and close all camera triangles within 0.30 frames; crops of the hands around the clap show contact about four frames earlier in the second camera than in the first. Read by frame index, pairs with the 29.60-fps camera are 20–27 frames off. The COLMAP calibration shipped with this capture places three cameras at the same centre, so 3D is not evaluated.

Semi-real drift. Webcam pair (host clock), 10 clips of 200 frames, E_{drift} for d = 0 / 1 / 2 / 3: CC-const 0.19 / 0.28 / 0.52 / 0.79; Caspi-style 0.46 / 0.44 / 0.40 / 0.44; CC-slide 0.42 / 0.46 / 0.55 / 0.56; CC-c2f 0.34 / 0.42 / 0.42 / 0.47; CSF-slide 0.36 / 0.25 / 0.33 / 0.61; CSF-c2f 0.35 / 0.27 / 0.32 / 0.39.

S9 Configuration by Regime

The per-regime summary behind the recommendations of the main paper (Discussion).

S10 Sequence-Held-Out Confirmation

Nine TotalCapture sequences of the official training partition, of subjects S1–S3 that also appear in the development and test data (so sequences, not subjects, are held out), were never used for development (walking1, acting1, freestyle1) or testing (walking2, acting3, freestyle3): S1–S3 × {walking3, acting2, freestyle2}. Their keypoints were extracted with the identical pipeline after all paper results were fixed, and every analysis was run once with the paper's commands; a claim counts as confirmed if its effect has the same sign (Table S11).

Table S11: Sequence-held-out confirmation on nine unused TotalCapture sequences (10-s clips, 5,022 samples): mean error in frames, paired differences with 95% bootstrap intervals over sequences, and the test-split value of the paper.

Claim	Held-out	Test (paper)
signed input, CC	1.46 → 0.58 (−0.88 [−1.42, −0.40])	1.32 → 0.48
signed input, CSF	1.01 → 0.47 (−0.55 [−0.81, −0.32])	0.97 → 0.43
CSF vs. CC	−0.11 [−0.31, +0.01]	−0.05
DTW / Caspi-style vs. CC	+0.13 / +0.19	+0.18 / +0.23
body vs. image axis (CC)	+0.03 [−0.03, +0.10]	+0.04
router δ=3	0.47 at 4% CSF; vs. CSF +0.01 [−0.04, +0.06]	0.43 at 4%
router δ=0.75	0.44 at 28%; vs. CC −0.14 [−0.32, −0.02], vs. CSF −0.02 [−0.05, −0.00]	0.41 at 25%
gate q=0.95 (CC)	kept 59%, precision 88→89%, walking kept 20%	60%, 91→96%, 17%
placement: tree / LS / gated LS	0.83 / 0.41 / 0.42 (75% placed)	0.59 / 0.34 / 0.32 (81%)
3D at d=3: none / single / CC-c2f / CSF-c2f / true	42.3 / 19.4 / 10.6 / 10.1 / 2.6 mm	44.3 / 18.0 / 9.7 / 9.0 / 2.0 mm

Table S12: Self-calibrated gate on TotalCapture 10-s test clips (pairwise CC estimates, signed input): share kept, precision (share of kept estimates within one frame), share of failures removed and mean error of kept estimates.

Gate	kept	precision	failures removed	error	walking kept
none	100%	91%	0%	0.49	100%
null q=0.90	64%	96%	68%	0.34	23%
null q=0.95	60%	96%	73%	0.32	17%
null q=0.99	52%	97%	81%	0.30	11%

Table S13: Timing behaviour of our five devices (container timestamps for phones, host timestamps for webcams).

Device	Recording	fps	Observation
Phone A (iPhone 7 Plus)	1080p HEVC	30.001	reference
Phone B (iPhone 12 Pro)	1080p HEVC, HDR	29.986	irregular intervals
Phone C (iPhone)	720p H.264	30.016	1 dropped frame
Webcam 0	720p, host timestamps	29.04	file claims 30 fps
Webcam 1	720p, host timestamps	29.99	—

All signs hold except one: gated least squares is not better than least squares over all edges on these sequences (0.42 vs. 0.41 frames), and the gate's precision gain is small (+1 point instead of +5). Absolute errors are about 0.1 frame higher than on the test split. The CSF axis variant is fixed by its checkpoint, so only CC compares the two axes here.

S11 Further Figures and Tables

Moved from the main paper for space: typical real pairs (Fig. S2; on a wide AthletePose3D pair the speed profile peaks five frames off and the signed one at the true lag, on walking the speed peak is shifted by part of a step, and on SportsPose a near-orthogonal pair is fixed while another, with the throwing arm edge-on in one view, still fails with both inputs), the gate at three quantiles (Table S12) and the devices of our first capture (Table S13).

S12 Reproducibility

Code (MIT license), the trained CSF checkpoints, the logs and summaries of every run reported here, our first two five-device captures (2D keypoints, device timestamps, calibration and the torch ground truth; the videos show the performers and are not released) and the capture tools are available at https://anonymous.4open.science/r/v-synclock/. The licenses of AthletePose3D, TotalCapture and SportsPose forbid redistributing their data; the repository regenerates every input from the original downloads (Colab notebooks for AthletePose3D and TotalCapture keypoints, `sp_extract_pose.py` for SportsPose). A Dockerfile with pinned dependencies and `smoke_test.py` reproduce the results of our second capture from the shipped data in a few minutes, without any licensed data. All real-image and Panoptic results of the paper are produced by `run_vy_suite.py --tag final` (resumable jobs; about two hours on a laptop CPU), the speed baselines by `--tag base`, the signal comparison by `signal_baselines.py`, roll by `roll_robustness.py`, the router by `tc_hybrid2.py`, streaming by `tc_online_rig.py`, cost by `rig_profile.py`, the captures by `take2_eval.py`, `take2_rig.py`, `take2_index.py` (reading by frame index) and `prev_raw_eda.py` (third capture; per-frame container times by `mov_timing.py`), router transfer by `short_router.py`, the sequence-held-out confirmation by `run_confirm.sh` with the notebook `totalcapture_confirm_colab.ipynb`, and the figures by `make_v2_figures.py`. Paired bootstrap intervals use 10,000 resamples with fixed seeds. The repository is prepared for ACM artifact review: the Docker image and smoke test run without licensed data, and every table is regenerated by the scripts above.

References

[1] David Pagnon, Mathieu Domalain, and Lionel Reveret. 2022. Pose2Sim: An open-source Python package for multiview markerless kinematics. *Journal of Open Source Software* 7, 77 (2022), 4362. doi:10.21105/joss.04362

[2] Scott D. Uhlrich, Antoine Falisse, Łukasz Kidziński, Julie Muccini, Michael Ko, Akshay S. Chaudhari, Jennifer L. Hicks, and Scott L. Delp. 2023. OpenCap: Human movement dynamics from smartphone videos. *PLOS Computational Biology* 19, 10 (2023), e1011462. doi:10.1371/journal.pcbi.1011462

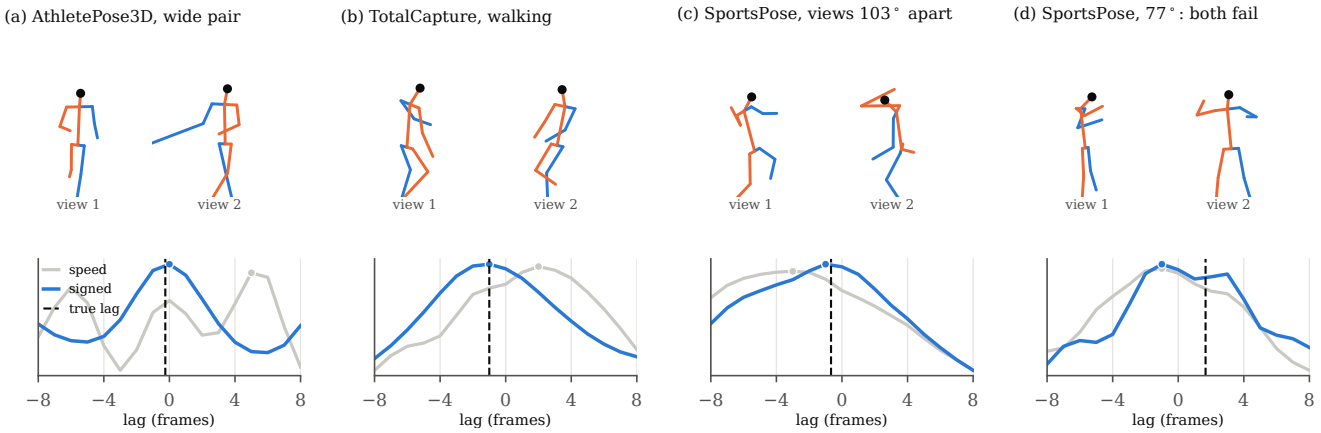


Figure S2: Real camera pairs with known offsets: detected 2D skeletons of both views at one instant (blue: left limbs) and the CC lag profiles with the speed input (grey) and the signed input (blue); dots mark the peaks, the dashed line the true lag. Each example is the median-ranked sample of its category (speed error >1.5 frames; signed error <0.25 frames, or >1.5 frames in (d)).